

CAPSTONE PROJECT 1

PIPELINED WALLACE TREE MULTIPLIER

Design. Verify. Optimize.

1. PROJECT OVERVIEW

Design and implement an **N-bit Pipelined Wallace Tree Multiplier** using Verilog HDL.

The design must use Wallace Tree reduction with pipelining to improve speed/throughput. Verify the design using a self-checking testbench.

2. OBJECTIVES

- Understand Wallace Tree multiplier architecture
- Implement compressor tree using Full Adders and Half Adders
- Insert pipeline registers at appropriate stages
- Verify functionality using testbench
- Analyze latency and throughput

3. DESIGN SPECIFICATIONS

Parameter	Specification
Operand Width (N)	8-bit (A[7:0] x B[7:0])
Output Width	16-bit
Architecture	Pipelined Wallace Tree
Pipeline Stages	3 Stages (configurable)
Clock	50 MHz (for simulation)
Language	Verilog HDL
Reset	Active Low Synchronous
Throughput	1 result per clock cycle after pipeline fill

7. TESTBENCH REQUIREMENTS

- Self-checking testbench
- Random test cases for all possible combinations (at least 1000 random cases)
- Check result after pipeline latency
- Display pass/fail summary
- Generate waveform

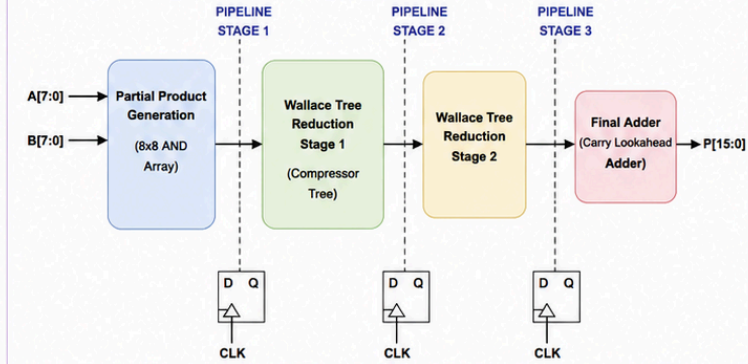
9. DELIVERABLES

- Verilog RTL Code
- Testbench (Self-checking)
- Simulation Waveform (Screenshot)
- Project Report (PDF)
- Design Documentation

10. EVALUATION CRITERIA

Functionality	40%
Code Quality	20%
Testbench Coverage	20%
Documentation	10%
Timely Submission	10%

4. BLOCK DIAGRAM



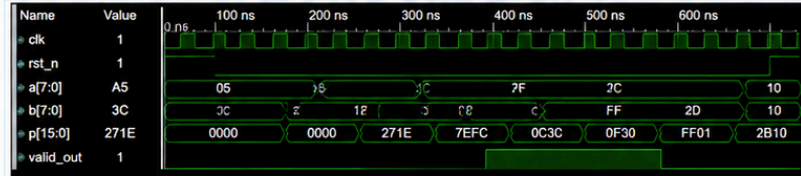
5. ARCHITECTURE DETAILS

- Partial Product Generation:** Generate all partial products using AND gates.
- Wallace Tree Reduction:** Use Full Adders (FA) and Half Adders (HA) to reduce partial products to two rows.
- Final Addition:** Add the two rows using Carry Lookahead Adder (CLA) or Ripple Carry Adder.
- Pipelining:** Insert registers between major stages to improve performance.

6. PIPELINE STAGES (EXAMPLE)

Stage	Function
Stage 1	Partial product generation + Register
Stage 2	Wallace Tree Reduction (Level 1) + Register
Stage 3	Wallace Tree Reduction (Level 2) + Register
Stage 4	Final Adder + Output Register

8. SIMULATION WAVEFORM (SAMPLE)



Note: Output P becomes valid after 3 clock cycles (pipeline latency).

11. SUBMISSION DETAILS

- Repository (GitHub Classroom): <https://classroom.github.com/a/xxxxxx>
- Branch: main
- Commit Message: "Capstone1_Final"
- Submission Date: **7 Days from Project Release Date (by 11:59 PM)**

NOTE

Follow good coding practices. Use meaningful module and signal names. Ensure your design is synthesizable and simulation clean.

All the best! Build it. Verify it. Optimize it! 🚀